

GCSE algebra - Set 2 - Solutions

Specification M4.1-M4.8, M4.18, M4.19.

Answer key

Question	Answer	Topic
1	C	GCSE algebra
2	A	GCSE algebra
3	E	GCSE algebra

Worked solutions

Question 1 ALG-15 · GCSE algebra · spec M4.18

The distance-time graph of a journey is a straight line from (0, 0) to (2, 60), then horizontal to (3, 60), then a straight line to (5, 0) (time in hours, distance from home in km). What was the average speed for the whole journey?

Answer: C

Total distance $60 + 60 = 120$ km in 5 hours: 24 km/h. (Displacement is 0 but distance travelled is not.)

Full method - how and why

Average speed uses **total distance**, not net displacement - and this journey returns home.

- 1 Read the outward distance: the traveller goes from 0 to 60 km, so 60 km out.
why: The graph shows distance **from home** against time.
- 2 The horizontal section is a stop (no distance covered), and the final section returns to 0: another 60 km.
why: Coming back still counts as distance travelled.
- 3 Total distance $= 60 + 60 = 120$ km over a total time of 5 hours.
why: Include the stopped hour in the time - it is part of the journey.
- 4 Average speed $= \frac{120}{5} = 24$ km/h.
why: Total distance divided by total time.

Where it goes wrong

- Using the net displacement 0 and answering 0 km/h - an offered option.
- Excluding the stopped hour, giving 30 km/h.

Takeaway Average speed = total distance $\div$ total time; stops count in the time, return legs count in the distance.

Question 2 ALG-03 · GCSE algebra · spec M4.6

Simplify $\frac{x^2 - 9}{x^2 + x - 12} \times \frac{x^2 - 16}{x^2 - 3x}$.

Answer: A

Factorise everything: $\frac{(x-3)(x+3)}{(x+3)(x-4)} \times \frac{(x-4)(x+4)}{x(x-3)}$. The $(x+4)$ factors cancel and one $(x-3)$ cancels, leaving $\frac{x+3}{x(x-3)}$.

Full method - how and why

Multiplying algebraic fractions: factorise everything first, then cancel common factors across the product.

- 1 Factorise each part: $x^2 - 9 = (x-3)(x+3)$; $x^2 + x - 12 = (x+4)(x-3)$; $x^2 - 16 = (x-4)(x+4)$; $x^2 - 3x = x(x-3)$.
why: Nothing can be cancelled until everything is in factor form.
- 2 Write the product: $\frac{(x-3)(x+3)}{(x+4)(x-3)} \times \frac{(x-4)(x+4)}{x(x-3)}$.
why: In a product of fractions any top factor can cancel with any bottom factor.
- 3 Cancel: the $(x+4)$ pair, and one $(x-3)$ from the first numerator with the first denominator.
why: There are two $(x-3)$ s on the bottom but only one on top, so one survives below.
- 4 What remains: $\frac{(x+3)(x-4)}{x(x-3)}$.
why: Count the leftovers carefully - this is where marks are lost.

Where it goes wrong

- Cancelling both $(x-3)$ factors and losing one from the denominator.
- Cancelling terms rather than factors, e.g. striking out the x^2 s.

Takeaway Factorise fully, then cancel factor-by-factor and count how many copies of each remain.

Question 3 ALG-08 · GCSE algebra · spec M4.19

The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?

- I The 5th term is 2.6.
- II Every term after the first is greater than the previous term.
- III Some term of the sequence equals 10.

Answer: E

I: $\frac{26}{10} = 2.6$. II: $u_n = \frac{n}{2} + \frac{1}{2n}$; $u_1 = 1$, $u_2 = 1.25$, and for $n \geq 1$ the increase $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$, so increasing. III: $n^2 - 20n + 1 = 0$ has no integer solutions (discriminant 396 is not a square), so false.

Full method - how and why

Three claims about a sequence. Splitting the formula makes the behaviour transparent.

- 1 I: $u_5 = \frac{25 + 1}{10} = \frac{26}{10} = 2.6$. **True.**

why: Direct substitution.

- 2 Split the formula: $u_n = \frac{n^2 + 1}{2n} = \frac{n}{2} + \frac{1}{2n}$.

why: Splitting shows one growing part and one shrinking part - much easier to reason about.

- 3 II: $u_{n+1} - u_n = \frac{1}{2} - \frac{1}{2n(n+1)}$, which is positive for all $n \geq 1$ since $\frac{1}{2n(n+1)} \leq \frac{1}{4}$. **True.**

why: The $\frac{n}{2}$ term grows by $\frac{1}{2}$ each step, while the $\frac{1}{2n}$ term shrinks by less than that.

- 4 III: solve $\frac{n^2 + 1}{2n} = 10 \Rightarrow n^2 - 20n + 1 = 0$, whose discriminant $400 - 4 = 396$ is not a perfect square, so n is not an integer. **False.**

why: Terms exist only for whole-number n , so a non-integer solution means no term equals 10.

- 5 I and II only.

why: Match to the options.

Where it goes wrong

- Accepting III because the equation has real solutions - they must be **positive integers**.
- Testing II on only the first two terms.

Takeaway Split a rational formula into simpler pieces; and remember sequence indices must be whole numbers.